Functional Specifications Document

Curator: Threaded Discussion and Schedule Platform

In Compliance with the Course:

Application Development and Emerging Technologies

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Document Control

Version	Date	Author	Description
1.0	October 28, 2026	Jamere Leventh A. Jalali; Kenneth I. Velasco	Initial draft for proposal submission
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Introduction

2.1 Purpose

This document specifies the functional requirements of Curator, a web application in which each class has its own server: a verified space for subject-based discussion, official announcements, and a shared schedule. It consolidates class communication that is currently spread across general-purpose chat groups. The application is designed for any school. The pilot configuration uses the institutional email domain of the Polytechnic University of the Philippines (@iskolarngbayan.pup.edu.ph), and the domain is configurable per school.

2.2 Scope

In scope: server creation with administrator confirmation; student enrollment verified by Certificate of Registration (CoR); subject channels with role-based access; a threaded forum with collapsible replies and a per-post choice of anonymous or named display; tagged announcements that generate a read-only class calendar; in-app notifications; moderation with logged authority; and administration.

Out of scope: real-time chat and voice or video calls; file storage beyond small attachments (learning materials are shared by external link); grade or attendance management; integration with an institution's official systems; native mobile applications (the web interface is mobile-first and responsive); email notifications (email is used only for account verification and password reset); professor accounts; and two-factor authentication.

Deferred to later versions: replies to replies (nested replies); muting and suspension of members; polls; a recurring weekly class schedule; calendar export, subject filters, and colors; automatic profanity filtering; Suggestions and Appeals channels; per-subject officer assignment and Vice President succession; officer requests for identity views; thread and learning-material pinning; reply stars; search and filtering; photo and file attachments; self-reveal of anonymous posts; channel muting and quiet hours; announcement view counts; password reset; the first-time walkthrough; screenshot evidence in abuse reports; and the Admin statistics dashboard.

2.3 Definitions

Term	Meaning
CoR	Certificate of Registration issued by the institution, listing a student's enrolled subjects and section
Server	A class space created by a President for one class and section, containing its channels and members
Channel	A space within a server: Announcements, Calendar, Rules, or a forum channel such as one subject
Thread	A top-level forum post, normally a question or topic, with its replies
Tag	A label on an announcement (for example Exam or Deadline) that sets its entry format and, for dated tags, creates a calendar event
Officer	A class officer, including the Vice President and subject-assigned beadles, who may post announcements and moderate
Beadle	An officer designated to one subject, responsible for its announcements and for communication with the professor
President	The member who creates and administers the server
Admin	The administrator who confirms server creation, manages accounts, and reviews reports of abuse of authority
Anonymous Label	A pseudonym of the form CSS-##### shown in place of a name and fixed within one thread
FSD	Functional Specifications Document
MVC	Model-View-Controller

2.4 References

- Republic Act No. 10173, Data Privacy Act of 2012
- Course materials, Application Development and Emerging Technologies

Overall Description

3.1 Problem Statement

Class information is typically shared through group chats where announcements are buried under unrelated messages. Students miss notices about class suspensions, room changes, or online sessions. Academic questions are lost in message streams, and many students hesitate to ask questions publicly for fear of being judged. Schedules are kept informally and are often inconsistent between students.

3.2 Proposed Solution

Curator separates the three needs that group chats combine. A President creates a server for the class once the Admin confirms the President's CoR, and students join with a CoR that the President verifies. Announcements are tagged and structured, and dated announcements are placed automatically on a shared calendar. Academic discussion happens in threaded, per-subject channels where the question is the focus and replies are collapsible. Before posting, each member chooses to appear anonymously or by name. The President and Admin can view the author of an anonymous reply under a disclosed, logged procedure, and members can report abuse of authority to the Admin.

3.3 Objectives

1. Ensure that only enrolled students access a subject's channel.
2. Make class announcements clear, structured, and impossible to miss.
3. Keep a single, consistent class schedule generated from tagged announcements.
4. Encourage students to ask questions without fear of judgment while preserving accountability, and keep the authority of officers itself accountable.
5. Display the professor's name and official work contact details in each subject channel.

3.4 Target Users

User	Description
Student	Reads announcements and the calendar, posts questions, replies, and may join several servers
Officer	Posts announcements for assigned subjects and moderates; includes the Vice President and subject beadles
President	Creates and administers the server, verifies join requests, assigns roles, and holds authorship-view authority
Admin	A department office or student council representative who confirms server creation, manages accounts, reviews reports of abuse of authority, and may delete servers

3.5 Assumptions and Constraints

- Every student possesses a CoR for the current term.
- The Admin is available to confirm server requests, and the President is available to verify join requests.
- Professor contact details are official work information entered by officers or the President and confirmed against official sources.
- CoR files contain personal data. Students block out the contact number before upload, and files are handled under the privacy rules in Section 11.

User Roles and Permissions

Function	Student	Officer	President	Admin
Register an account	Yes	Yes	Yes	No
Create a server (CoR confirmed by Admin)	Yes	Yes	Yes	No
Confirm server requests; delete a server	No	No	No	Yes
Join a server (CoR verified by President)	Yes	Yes	No	No
Verify join requests	No	No	Yes	No
Change a member's channel access	No	Yes	Yes	No
Assign officer roles	No	No	Yes	No
Transfer the President title	No	No	Yes	No
Add forum channels; edit professor details	No	Yes	Yes	No
View permitted channels	Yes	Yes	Yes	Reported servers only
Post a thread or reply (anonymous or named)	Yes	Yes	Yes	No
View reply authorship (reason required, logged, author notified)	No	No	Own server	Reported servers
Post announcements	No	Yes	Yes	No
Hide or remove content (reason shown to owner)	No	Yes	Yes	Yes
Report a thread or reply	Yes	Yes	Yes	No
Report abuse of authority to the Admin	Yes	Yes	Yes	Reviews
Review the audit log	No	No	Own server	Yes
Manage accounts	No	No	No	Yes

All officers hold the same authority. If a President cannot transfer the title, the Admin reassigns it. The Admin views reply authorship only in servers named in an abuse report.

Themes and User Stories

The requirements are organized into six themes, in priority order: Authentication and roles; CoR registration and servers; Threaded discussion; Announcements and calendar; Anonymity and accountability; and Moderation and administration. Each story is mapped to its functional requirements in Section 6.

ID	Theme	Story	Requirements
US-01	Authentication and roles	As a student, I want to create an account and verify my school email, so that my identity is confirmed.	FR-001 to FR-003
US-02	CoR registration and servers	As a student, I want to request a server for my class by submitting my CoR, so that the Admin can confirm that I may create it.	FR-004, FR-005
US-03	CoR registration and servers	As a student, I want to submit my CoR, with the contact number blocked out, and my details, so that I get access to my subjects only.	FR-006, FR-007
US-04	CoR registration and servers	As the President, I want to approve or reject join requests with a reason, so that only enrolled students join.	FR-008, FR-009
US-05	CoR registration and servers	As a student, I want to see only the channels of my subjects, so that I am not distracted by others.	FR-010, FR-016
US-06	CoR registration and servers	As a student, I want to see my professor’s name and official contact details in each subject channel, so that I know how to reach them.	FR-015

ID	Theme	Story	Requirements

US-07	CoR registration and servers	As the President, I want to add subject channels and assign officers, so that the server matches my class.	FR-012, FR-014, FR-015
US-08	CoR registration and servers	As the President, I want to change a member's channel access and transfer the President title when needed, so that membership stays accurate.	FR-011, FR-013
US-09	Threaded discussion	As a student, I want to post a question with a title and details, so that classmates can help.	FR-018
US-10	Threaded discussion	As a student, I want to scan a list of questions with replies hidden and open one to read its replies, so that I read only what I need.	FR-020, FR-021
US-11	Threaded discussion	As a student, I want to reply to a question and edit my own posts, so that I can help and fix mistakes.	FR-022, FR-023
US-12	Threaded discussion	As a student, I want to choose before posting whether to appear anonymously or by name, so that I am not afraid of being judged.	FR-019, FR-025
US-13	Announcements and calendar	As an officer, I want to post a tagged announcement (for example Class Status, Exam, or Deadline), so that the class is clearly informed.	FR-032, FR-033
US-14	Announcements and calendar	As a student, I want announcements grouped by subject and a notification for each new one, so that I do not miss them.	FR-035, FR-038
US-15	Announcements and calendar	As a student, I want dated announcements to appear on a read-only calendar and update when edited or cancelled, so that I have one reliable schedule.	FR-034, FR-036, FR-037
US-16	Anonymity and accountability	As a student, I want to read the class rules and be told in advance that the President can view authorship, so that I understand the rules.	FR-017, FR-028

ID	Theme	Story	Requirements
US-07	CoR registration and servers	As the President, I want to add subject channels and assign officers, so that the server matches my class.	FR-012, FR-014, FR-015
US-17	Anonymity and accountability	As the President, I want to view who wrote a reported anonymous reply after stating a reason, so that abuse can be addressed; the view is logged and the author is notified.	FR-024, FR-026, FR-027, FR-042
US-18	Anonymity and accountability	As the Admin, I want to review the audit log, so that misuse of authorship views can be detected.	FR-029
US-19	Anonymity and accountability	As a student, I want to report abuse of authority by an officer or the President to the Admin, so that those with authority can be held accountable.	FR-030, FR-031
US-20	Moderation and administration	As a student, I want to report an inappropriate thread or reply, so that moderators can act.	FR-039
US-21	Moderation and administration	As the President or an officer, I want to hide or remove violating content with a reason, so that channels stay respectful.	FR-040
US-22	Moderation and administration	As the Admin, I want to deactivate accounts and delete servers, so that misuse can be stopped.	FR-041

Functional Requirements

Priority: M = Must have, S = Should have. Could-have items are deferred (Section 2.2).

6.1 Account and Authentication

ID	Requirement	Priority
FR-001	The system shall allow a user to create an account with an institutional email address (domain configurable per school) verified by emailed link.	M
FR-002	The system shall authenticate users by email and password with hashed storage; passwords shall have at least 8 characters with uppercase, lowercase, and a digit, and may include special characters.	M
FR-003	The system shall show the rules and privacy notice at registration and require consent to CoR processing and authorship visibility.	M

6.2 Servers and Membership

ID	Requirement	Priority
FR-004	The system shall allow a student to request a server by submitting a CoR (PDF, up to 6 MB) to the Admin.	M
FR-005	The Admin shall approve or reject a server request with a reason; approval creates the server with its default channels and makes the requester its President.	M
FR-006	The system shall allow a student to request to join a server by submitting a CoR (PDF, up to 6 MB) and confirming name, student number, and the subjects and sections for that server.	M
FR-007	The submission screen shall instruct the student to block out the contact number before upload, and the file shall be visible only to the verifier.	M
FR-008	The President shall approve or reject join requests with a reason (for example unreadable file, name mismatch, wrong term).	M

ID	Requirement	Priority
FR-009	On approval the system shall record the membership, grant access to the channels of the confirmed subjects, and delete the stored CoR file.	M
FR-010	A student with subjects in different sections shall be able to join several servers and see only the subject channels taken in each.	S
FR-011	The President and officers shall be able to change a member's channel access at any time, effective immediately.	S
FR-012	The President shall be able to assign and remove officer roles.	M
FR-013	The President shall be able to transfer the President title to another member.	S

6.3 Channels

ID	Requirement	Priority
FR-014	Each server shall include the default channels Announcements, Calendar, and Rules.	M
FR-015	The President and officers shall add and manage subject channels and other forum channels; each subject channel header shall show the professor's name and official work contact details.	M
FR-016	The system shall show a member only the channels the member may access.	M
FR-017	The Rules channel shall state the class rules, the powers of each role, and the consequences of breaking rules.	M

6.4 Threaded Discussion

ID	Requirement	Priority
FR-018	The system shall allow a member to post a thread with a title and body, with text formatting and standard length limits.	M
FR-019	Before posting a thread or reply, the member shall choose	M

ID	Requirement	Priority
	anonymous or named display.	
FR-020	The channel view shall list threads showing title, author display, reply count, and last activity, with replies hidden by default.	M
FR-021	The system shall allow a member to open a thread and expand or collapse its replies.	M
FR-022	The system shall allow members to reply to a thread; replies appear in a single collapsible list under the thread.	M
FR-023	The system shall allow authors to edit their own threads and replies in place; removal is done by the President or officers.	M

6.5 Anonymity and Accountability

ID	Requirement	Priority
FR-024	The system shall store the true author of every thread and reply.	M
FR-025	Anonymous authors shall appear under a label of the form CSS-##### fixed within a thread; replies by the thread's poster shall also carry an "Author" label.	M
FR-026	The President (in their server) and the Admin (in servers named in an abuse report) shall be able to view the name and section of an anonymous author after stating a reason, and each view shall be recorded in the audit log (viewer, post, reason, time).	M
FR-027	The system shall notify the author when their identity is viewed.	M
FR-028	The system shall inform students, at registration and when choosing anonymity, that the President and Admin can view authorship.	M
FR-029	The President and Admin shall be able to review the audit log.	M
FR-030	A member shall be able to report abuse of authority by an officer or the President to the Admin with a written description.	S
FR-031	The Admin shall be able to view the channels of a server named	S

ID	Requirement	Priority
	in an abuse report.	

6.6 Announcements and Calendar

ID	Requirement	Priority
FR-032	Officers and the President shall be able to post an announcement, selecting a subject and a tag; dated tags shall require a date and time.	M
FR-033	Tags shall be Class Status (held, not held, or rescheduled), Assignment, Quiz, Exam, Activity, Deadline, Event, Learning Material, and General; all tags share one entry form with title, body, and optional date, time, and room or link.	M
FR-034	Announcements shall publish immediately and be editable, with an “Edited” label.	M
FR-035	The Announcements channel shall group announcements by subject, and each announcement shall also appear in its subject channel.	M
FR-036	The system shall create a calendar event from each announcement with a dated tag, update it when the announcement is edited, and mark it cancelled when the class is marked not held.	M
FR-037	The calendar shall be read-only; events change only through announcements.	M

6.7 Notifications

ID	Requirement	Priority
FR-038	The system shall send in-app notifications for new or edited announcements, replies to a user’s threads or replies, and moderation outcomes including authorship views.	M

6.8 Moderation

ID	Requirement	Priority
FR-039	The system shall allow members to report a thread or reply with a reason: cheating, harassment, spam, or bullying.	M
FR-040	The President and officers shall be able to hide or remove content, with the reason shown to its owner.	M

6.9 Administration

ID	Requirement	Priority
FR-041	The Admin shall manage accounts (deactivate and restore) and delete servers.	M
FR-042	The system shall keep one audit log of authorship views and moderation actions.	M

Key Use Cases

UC-01 Create a Server

- **Actor:** Student (future President), Admin
- **Main flow:** (1) The student submits a CoR and requests a server. (2) The Admin checks the CoR and approves. (3) The system creates the server with its default channels and makes the student its President. (4) The President adds subject channels and invites classmates.
- **Alternate flow:** If rejected, the Admin enters a reason and the student may resubmit.

UC-02 Join a Server

- **Actor:** Student, President
- **Main flow:** (1) The student creates an account and verifies the email. (2) The student selects the server, uploads the CoR with the contact number blocked out, and confirms name, student number, subjects, and sections. (3) The system queues the request for the President. (4) The President compares the CoR with the entered details and approves. (5) The system records the membership, grants access to the subject channels, and deletes the CoR file. (6) The student is notified.
- **Alternate flow:** If rejected, the President enters a reason and the student may resubmit.

UC-03 Post a Question

- **Actor:** Member
- **Precondition:** The member has access to the channel.
- **Main flow:** (1) The member selects New Thread in a subject channel. (2) The member enters a title and body and chooses anonymous or named display, after reading the authorship notice. (3) The system saves the thread and lists it in the channel.

UC-04 Reply to a Thread

- **Actor:** Member
- **Main flow:** (1) The member opens a thread and expands its replies. (2) The member writes a reply and chooses anonymous or named display. (3) The system saves the reply with its true author and shows the chosen name or an anonymous label.

UC-05 Post a Tagged Announcement

- **Actor:** Officer (assigned subject), President
- **Main flow:** (1) The officer selects New Announcement, a subject, and a tag. (2) The officer completes the entry format of the tag. (3) The system publishes the announcement in the Announcements channel and the subject channel. (4) For dated tags the system creates the calendar event. (5) The system notifies members.
- **Alternate flow:** If the announcement is edited, the event is updated; if the class is marked not held, the event is cancelled.

UC-06 View Reply Authorship

- **Actor:** President, Admin
- **Main flow:** (1) The actor selects a reported or flagged anonymous reply. (2) The system requires a stated reason. (3) The system displays the author's name and section and records the view in the audit log. (4) The system notifies the author.

UC-07 Report Abuse of Authority

- **Actor:** Member, Admin
- **Main flow:** (1) The member submits a report with a written description to the Admin. (2) The Admin reviews the named server's channels and audit log. (3) The Admin may remove content, suspend accounts, or delete the server.

Business Rules

- BR-01: A student can access a subject channel only if the verified CoR lists that subject for the server's section.
- BR-02: Anonymity applies to display only; it does not exempt a user from class rules or institutional policy.
- BR-03: Authorship may be viewed only by the President in their server and by the Admin in servers named in an abuse report; each view requires a stated reason, is logged, and notifies the author.
- BR-04: Professor details shown in a channel must be official work contact information only.
- BR-05: The Admin confirms server requests; the President verifies join requests. The Admin has no control over a member's channel access.
- BR-06: Hiding or removing content requires a reason that is shown to the affected member.
- BR-07: A deleted account appears as "Deleted account"; its content remains.

System Architecture and Data Model

9.1 MVC Structure

Layer	Responsibility	Examples
Model	Data entities, validation, database access	User, Server, Membership, Channel, Thread, Reply, Announcement, CalendarEvent, AuditLog
View	Pages and components shown to the user (Django templates)	Channel list, thread list, thread detail, calendar, announcement form
Controller	Request handling and business logic (Django views and URL routing)	Account, server, CoR request, channel, thread, announcement, calendar, moderation, and admin views

9.2 Proposed Technology Stack

Component	Proposed Choice
Back end	Python with the Django framework, whose Model-Template-View structure corresponds to MVC
Views	Django templates with responsive, mobile-first CSS in a dark theme and light client-side scripting for collapsible replies
Authentication and admin	Django's built-in authentication, with the Admin panel built on Django's admin interface
Database	PostgreSQL on Supabase, accessed through the Django ORM
File storage	Supabase Storage (private bucket) for temporary CoR files
Hosting	Render (web service)
Version control and backlog	Git and GitHub; GitHub Projects

9.3 Data Model

Entity	Key Attributes
User	user_id, email, password_hash, full_name, student_no, is_active, created_at
Server	server_id, name, school_domain, president_id, status, created_at
CoRRequest	request_id, request_type (server or join), user_id, server_id, confirmed_subjects, status, decided_by, decision_reason, decided_at
Membership	user_id, server_id, role
Channel	channel_id, server_id, type, name, subject_code, professor_name, professor_email, office_hours
ChannelAccess	user_id, channel_id
Thread	thread_id, channel_id, author_id, is_anonymous, anon_label, title, body, status, created_at, edited_at
Reply	reply_id, thread_id, author_id, is_anonymous, anon_label, body, status, created_at, edited_at
Announcement	announcement_id, channel_id, author_id, tag, class_status, date, start_time, end_time, room_or_link, title, body, created_at, edited_at
CalendarEvent	event_id, announcement_id, channel_id, start_at, end_at, status
Report	report_id, reporter_id, report_type, target_id, reason, description, status
Notification	notification_id, user_id, message, is_read, created_at
AuditLog	log_id, actor_id, action, target_type, target_id, reason, timestamp

9.4 Entity Relationships

- A Server has many Memberships and Channels; each Membership holds one role.
- A User may hold Memberships in many Servers, with ChannelAccess limited to the subjects confirmed in each.
- A CoRRequest belongs to a User and, for join requests, to a Server.
- A Channel has many Threads and Announcements.
- A Thread has many Replies.
- An Announcement with a dated tag has one CalendarEvent.
- An AuditLog entry records each authorship view and moderation action.

User Interface Specifications

Screen	Description	Roles
Login / Register	Account access, consent, and email verification	All
Create Server	CoR upload for Admin confirmation	Student
Join Server	CoR upload and confirmation of details	Student, Officer
Request Queue	Join requests with CoR preview (President); server requests (Admin)	President, Admin
Channel List	First screen after login: the user's servers and channels	Members
Announcements	Subject topics with tagged announcements and edited labels	Members; Officers and President post
Calendar	Read-only events	Members
Channel View	Header with professor details and thread list with replies hidden	Members
Thread Detail	Thread body and collapsible replies	Members
New Thread / Reply	Editor with anonymous or named choice and authorship notice	Members
Rules	Class rules, powers of each role, and consequences	Members
Moderation Panel	Reports, hidden content, and authorship view with reason prompt	President, Officers; Admin for reported servers
Member Management	Roles, officer assignments, and channel access	President, Officers
Admin Panel	Server requests, abuse reports, accounts, and audit log	Admin
Notifications	List of in-app notifications	All

Interface standards:

- Mobile-first responsive layout in English, in a formal tone. On phones the layout follows familiar messaging-app navigation; on desktop the channel list sits at the left with the forum area as the main panel.
- A single dark theme with a minimalist, flat style, using no official or trademarked school visuals.
- Thread titles as the primary element, replies collapsed by default, and clear tag labels on announcements.
- A confirmation prompt before deleting or removing content, viewing authorship, and submitting a CoR.
- Accessible color contrast and usability on low-bandwidth connections.

Non-Functional Requirements

Category	Requirement
Security	Hashed passwords, role-based access control, protection against SQL injection and XSS, anti-forgery tokens on forms, HTTPS in deployment, private storage for CoR files, and session expiry
Privacy	Collection limited to necessary data in line with RA 10173; consent at registration; CoR deleted after verification; users informed in plain language about data use and authorship visibility
Performance	Channel and calendar pages respond within 3 seconds under normal load; supports about 50 concurrent users in the demonstration and a server of 40 to 55 members
Usability	A member can submit a question within two minutes of entering a channel; mobile-first design
Reliability	Daily database backup; logs are not silently deleted
Maintainability	Code organized by the MVC pattern (Django's Model-Template-View), documented, and version controlled
Compatibility	Current versions of major desktop and mobile browsers
Data retention	Threads, announcements, and logs are retained; CoR files are deleted after the decision; a deleted account appears as "Deleted account"

Acceptance Criteria

The proposal is considered satisfied when:

1. A student can request a server, and the Admin's confirmation makes the student its President with the default channels.
2. A student with an approved CoR sees only the channels for their enrolled subjects.
3. An officer's tagged announcement creates a correct calendar event and notifies members.
4. Threads display with replies collapsed, and replies show the anonymous label or name the author chose.
5. Authorship can be viewed only by authorized roles, every view is logged, and the author is notified.
6. The application is deployed and accessible for review.

Approval

Name	Role	Signature	Date
Jamere Leventh A. Jalali	Developer, Product Owner		
Kenneth I. Velasco	Developer, Scrum Master		
Eugene Andrew Bato	Course Instructor		